

3(a)(i)	(human volunteer / person and) no (mosquito) repellent / spray or (human volunteer / person and) water (spray) ; or (human volunteer / person and) solvent (from repellent / spray) ;	1
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Question	Answer	Marks								
3(a)(ii)	named risk <u>and</u> matching precaution ;	1								
	<table><tr><th>risk</th><th>precaution</th></tr><tr><td>mosquitoes take blood meal (from scientist) or mosquitoes bite (scientist)</td><td>protective clothing / gloves / masks / PPE</td></tr><tr><td>mosquitoes transmit, disease / pathogen / yellow fever / AW</td><td>protective clothing / gloves / masks / PPE use, disease / pathogen, free mosquitoes vaccine (against yellow fever)</td></tr><tr><td>(mosquito) repellent / spray / solvent, allergy / irritant / toxic</td><td>protective clothing / gloves / masks / PPE</td></tr></table>		risk	precaution	mosquitoes take blood meal (from scientist) or mosquitoes bite (scientist)	protective clothing / gloves / masks / PPE	mosquitoes transmit, disease / pathogen / yellow fever / AW	protective clothing / gloves / masks / PPE use, disease / pathogen, free mosquitoes vaccine (against yellow fever)	(mosquito) repellent / spray / solvent, allergy / irritant / toxic	protective clothing / gloves / masks / PPE
	risk		precaution							
	mosquitoes take blood meal (from scientist) or mosquitoes bite (scientist)		protective clothing / gloves / masks / PPE							
	mosquitoes transmit, disease / pathogen / yellow fever / AW		protective clothing / gloves / masks / PPE use, disease / pathogen, free mosquitoes vaccine (against yellow fever)							
(mosquito) repellent / spray / solvent, allergy / irritant / toxic	protective clothing / gloves / masks / PPE									

Question	Answer	Marks
3(b)	any four from: 1 98% DEET is effective (at repelling mosquitoes) ; 2 <i>idea that</i> standard error / SE, overlap for 98% DEET and lemon eucalyptus oil, so these mosquito repellents are equally effective / no significant difference between these mosquito repellents ; 3 <i>idea that</i> lemon eucalyptus oil is, (slightly) more effective / equally effective (as 98% DEET), but, has a low concentration (of chemical) / might be safer or <i>idea that</i> lemon eucalyptus oil is, (slightly) more effective / equally effective, (as 98% DEET) but DEET is, toxic / irritant / harmful ; 4 <i>idea that</i> not true for all concentrations of DEET / 40% (concentration) DEET is much less effective ; ora 5 no statistical test carried out ; 6 only one human volunteer tested ; 7 <i>idea that</i> some mosquitoes are still attracted to the human volunteer (when using 98% DEET) ; 8 <i>idea that</i> mosquitoes used / <i>Aedes aegypti</i>, do not transmit malaria ; 9 <i>idea that</i> mosquitoes were not able to, take a blood meal from / bite, the human volunteer / investigation only measured attraction of mosquitoes (to the human volunteer) ;	4